

Discovery Philosophy Engineering. Making Philosophical Claims Experimentally Interrogable

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Abstract

Philosophy is rich in claims about which distinctions matter, which descriptions are equivalent, what counts as rational or objective, and which structures are fundamental. Many such claims stay insulated from empirical pressure because they are stated at a level too informal to generate a discriminating test. That invariance under admissible transformation marks what is real rather than how we happened to describe it is an old idea with four developed statements already, in the study of what a transformation group leaves fixed, in robustness as agreement across independent means of detection, in invariance under intervention as the mark of a causal relation, and in the structure that survives the replacement of one theory by another. This paper does not propose a fifth. It proposes that the idea be made to cost something, by compiling a philosophical claim into an executable invariance test, declaring the admissible class in advance as a versioned artifact, reducing each failure to the smallest transformation that produces it, naming the component that absorbs the change, and recording the revision that follows. The result is a discipline we call Discovery Philosophy Engineering, in which an implementation is not the endpoint of philosophical analysis but an instrument for experimenting on it. We define an invariance envelope as the family of transformations under which a claim survives, a boundary witness as the smallest admissible transformation that breaks it, and a comparative notion of fundamentality that ranks two descriptions of comparable adequacy by their envelopes. Against the standing objection that the theorist picks the admissible class, we argue that declaring it is not sufficient, since a recorded arbitrary choice is still arbitrary, and give five constraints that make the class costly to gerrymander, the last of which requires that any narrowing of a claim be shown to bind. We then report the discipline running. A campaign graded four of its own theory's predictions through registrations sealed before the data were read, returning one pass whose interesting number is the half of the population that violates the law, two indeterminate verdicts whose reduction located where a theorem stops applying, and one pass in which two of three declared budgets behaved as the theory said and the third did not. A discovery line asked one question nine times, failed three of the nine by clauses written in advance, and ended with one law, one declared scope, and two recorded boundaries. The check the discipline prescribes then found a defect in its own authors' records that reading them had not. What is new here is not a criterion of invariance. It is the accounting that lets somebody else test one.

Keywords: invariance, robustness, objectivity, scientific discovery, representation independence, theory revision, fundamentality, preregistration, computational philosophy

1 Introduction

Philosophical inquiry repeatedly asks questions of one form. Which distinctions are real? Which descriptions are different presentations of the same case? Which aspects of a judgment are objective,

and which depend on a standpoint? What is preserved when a problem is translated, reparameterized, rescaled, or embedded in a richer representation? And when two theories disagree, which disagreement reflects the world and which reflects the language in which the world was described?

These are questions about invariance, and philosophy has answered them in that vocabulary for a long time. Section 2 sets out four developed answers, because the argument here depends on conceding how much of its central idea is already held. What this paper adds is not another account of why invariance matters. It is a way of making a particular invariance claim cost something to assert.

The difficulty is not that philosophers lack a criterion. It is that a criterion stated in prose leaves three things in the claimant's hands and shows the reader none of them. The class of transformations said to be irrelevant is chosen by the theorist, and a class chosen after the result is a class fitted to it. A failure is reported as a failure, without the smaller failure inside it that would say which assumption gave way. A commitment revised in the light of a counterexample is revised silently, so the record cannot separate a theory that learned from a theory that was never wrong. None of these is a defect of the invariance idea. Each is a defect of the accounting around it.

Experimental philosophy has subjected philosophical judgments to empirical study and found both stability and fragility in methods of cases (Knobe and Nichols, 2024; Irikefe, 2025). Computational philosophy uses mechanized methods to extend philosophical analysis (Thagard, 2024). Formal philosophy turns conceptual commitments into mathematical structures. Automated scientific discovery searches data for compact laws, increasingly through symbolic regression and closed-loop systems (Makke and Chawla, 2024; Kramer et al., 2026). Each supplies a necessary piece. None supplies a discipline for taking one philosophical proposition, compiling it into invariance and boundary claims, declaring in advance what would refute it, attacking it, and recording what the attack did to it in a form a third party can check.

This paper proposes that discipline, which we call Discovery Philosophy Engineering. It extends Philosophy Engineering, a method for operationalizing philosophical commitments through declared equivalences, invariance requirements, enforcement mechanisms, and machine-checkable artifacts (Bond, 2026c). The extension closes the loop. Rather than moving only from a philosophical principle to a formal constraint to an implementation, the implementation is turned back on the principle as an instrument. If a proposition can be formalized well enough to implement, the implementation usually permits experiments on it that argument alone cannot perform.

The contribution is methodological. Many philosophical questions are conceptual, normative, or modal in ways no test settles, and the discipline claims nothing about those. It applies where a philosophical claim entails observable invariance, representation, discrimination, or boundary structure, and its aim is to sharpen those entailments until success and failure are both informative.

2 Invariance has been asked to do this work before

The thought that what survives admissible transformation is a better candidate for the real than what does not is old, and it has at least four developed statements. A proposal in this area earns attention only by saying what it adds to them.

Geometry as the study of what a group leaves fixed. The Erlangen reading classifies geometries by the transformation groups whose invariants they study (Klein, 1893), and the extension of that reading to physical law is Weyl (1952). This is the origin of the move and remains its clearest case. What it supplies is the form of the question. What it does not supply is guidance when the admissible group is not handed over by the subject matter, which is the normal situation

outside mathematics and physics.

Objectivity as invariance. Nozick (2001) makes invariance under admissible transformations the account of objectivity itself, so that an objective fact is one holding across the transformations that do not matter. This is the closest existing statement of the present thesis, and Definition 3 below is a near relative of it. Nozick offers an analysis of a concept. He does not offer a procedure for deciding a particular case, and the question of who fixes the admissible transformations is raised rather than answered.

Robustness as agreement across independent means of detection. Levins (1966) defended robustness analysis in model building and Wimsatt (1981) generalized it, using the word robust for exactly what is invariant across independent means of determination. This is the structure of the invariance envelope of Section 4.3, reached three decades earlier and from the practice of working scientists rather than from geometry. The difference is bookkeeping rather than concept. Robustness analysis reports that a result held across several independent routes. It does not generally declare the set of routes in advance, does not require failures to be reduced, and has no standard form in which a reader can see which routes were tried and rejected. The literature knows this. Orzack and Sober (1993) press the point against Levins directly, arguing that models called independent are frequently not, and Weisberg (2006) reconstructs robustness analysis so that what it delivers is a common causal structure across a set of models rather than confirmation, which restricts the method to what it can support at the cost of narrowing what it establishes. Hacking (1983) makes the nearest relative of the present argument, that agreement between detection methods resting on different physical principles is evidence the detected thing is not an artifact of either. All three turn on the same unstated quantity, which is the set of routes actually available and actually tried. The envelope is that set, written down before the trying.

Invariance under intervention. Woodward (2003) makes invariance under a range of interventions the mark separating a causal generalization from an accidental one, and treats invariance explicitly as a matter of degree and range rather than as a predicate. The degree and range treatment is the move made here with the envelope, and Woodward’s account is the more developed of the two as philosophy. What it is not is a procedure with an audit trail, since the range over which a relation is invariant is established by argument and by experiment reported in the ordinary way.

Structure as what survives theory change. Structural realism holds that what carries across a scientific revolution is mathematical structure rather than ontology (Worrall, 1989). Replace the transformation from one theory to its successor by a transformation from one representation to another and the resemblance is close. Ismael and van Fraassen (2003) sharpen the related point that a symmetry marks superfluous theoretical structure, which is the philosophical form of the factorization criterion stated as Proposition 1 below. Their argument and that proposition say one thing in two vocabularies.

The concession is therefore large. The criterion is not new, the treatment of invariance as a matter of degree is not new, the use of a symmetry to identify structure that can be quotiented away is not new, and the identification of objectivity with invariance is not new. Five of this paper’s central ideas have owners.

What none of them supplies is an account under which an invariance claim can be checked by somebody who does not trust the claimant. In each, the admissible class is settled inside the argument that uses it. That is not a criticism of the authors, because prose was the medium they had. It is a description of what is left to be added. The proposal here is that the three things a prose statement leaves implicit, the class, the reduction of the failure, and the revision, become artifacts declared before the test, fixed by content hash, and refused by a checker when absent. Mayo (2018) supplies the notion that makes this worth doing, since a test is severe when the claim would probably have failed it had the claim been false, and severity is exactly what a class chosen

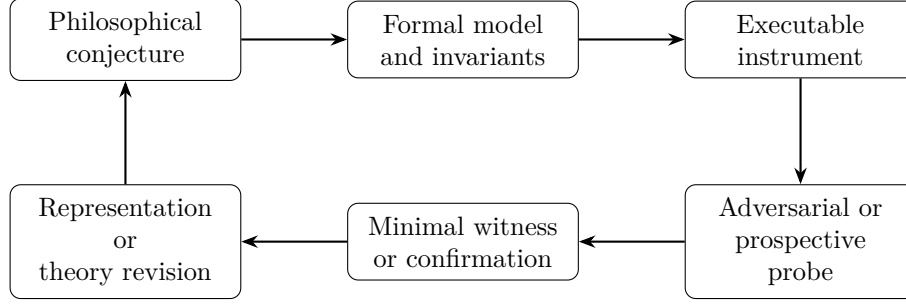


Figure 1: The loop the discipline runs. The arrow that separates it from ordinary hypothesis testing is the one from witness to revision, since a counterexample that has been reduced says which assumption must give way and an unreduced one does not.

after the fact destroys.

The novelty claimed here is accordingly narrow. It is not a new criterion of invariance. It is the accounting that lets an old one be operated by more than one person.

3 A claim becomes an instrument when its equivalences are declared

Philosophy Engineering was introduced as the systematic translation of epistemic, ethical, and logical commitments into computationally enforceable constraints on autonomous systems (Bond, 2026c). Its procedure has four steps. Declare a domain-specific equivalence relation over representations. Declare what must remain invariant. Implement canonicalization, quotienting, equivariance, or another enforcement mechanism. Package transformation trials and witnesses as machine-checkable artifacts. Its central example is the Epistemic Invariance Principle, which holds that a judgment procedure should not change its conclusion under transformations declared to preserve task-relevant meaning.

Discovery Philosophy Engineering generalizes this in the reverse direction. A philosophical claim becomes a source of formal commitments that a transformation grammar and an observation process can interrogate. The engineering artifact stops being the endpoint of analysis and becomes an instrument for experimenting on it.

Principle 1 (Discovery Philosophy Engineering). *A philosophical conjecture is a candidate for Discovery Philosophy Engineering when it can be translated into at least one non-trivial formal constraint whose satisfaction or violation is observable under a declared family of transformations, interventions, or representation changes.*

Non-triviality is essential. A constant judgment rule is invariant under every transformation and has learned nothing. Philosophy Engineering therefore pairs invariance with discriminative adequacy, so that irrelevant differences must not matter and relevant differences must be allowed to matter, and the present discipline inherits that dual obligation. Figure 1 shows the loop.

4 Formal architecture

4.1 Representations, transformations, and judgments

Let S denote a space of world states or cases, X a representation space, and $\sigma : S \rightarrow X$ an encoding map. Let Y be an output space, and let a judgment or theory-induced prediction be

$$J : X \rightarrow \Delta(Y), \quad (1)$$

where $\Delta(Y)$ is the space of distributions over Y and a deterministic judgment is a degenerate distribution. Let $\mathcal{T}$ be a set of transformations $\tau : X \rightarrow X$ declared to preserve some target-relevant structure.

The declaration matters, and it is where this architecture departs from the accounts of Section 2. No method discovers relevance without assumptions. An admissible transformation is not anything that can be done to the data. It is a transformation licensed by the philosophical claim under test. Unit conversion, variable renaming, permutation of independent premises, paraphrase preserving formal denotation, change of basis, population transfer, and stake rescaling are different transformations with different semantics, and a claim surviving one says nothing about another.

Definition 1 (Invariance claim). *Given an output equivalence relation $\sim_Y$, the judgment J is invariant under $\mathcal{T}$ if*

$$J(x) \sim_Y J(\tau x) \quad \forall x \in X, \tau \in \langle \mathcal{T} \rangle, \quad (2)$$

where $\langle \mathcal{T} \rangle$ is the closure generated by $\mathcal{T}$ when composition is meaningful.

This is the formal core of the Epistemic Invariance Principle (Bond, 2026c). Read as a compliance property it is a specification. Read as this paper reads it, it is an empirically attackable consequence of the philosophical commitment that the transformations in $\mathcal{T}$ do not matter.

4.2 A theory calling a difference representational should factor through the quotient

Let $x \sim_{\mathcal{T}} x'$ when two representations lie in the same orbit under the admissible transformations. The quotient $X / \sim_{\mathcal{T}}$ identifies every description the theory says is the same case.

Proposition 1 (Factorization criterion). *If J is constant on every $\sim_{\mathcal{T}}$ equivalence class, then there exists a well-defined map $\bar{J} : X / \sim_{\mathcal{T}} \rightarrow \Delta(Y)$ such that $J = \bar{J} \circ q$, where $q : X \rightarrow X / \sim_{\mathcal{T}}$ is the quotient map.*

Proof. Define $\bar{J}([x]) = J(x)$. Class constancy makes the definition independent of the representative chosen. Then $\bar{J}(q(x)) = J(x)$ by construction. $\square$

The mathematics is elementary and the reading carries the weight. A theory calling certain differences merely representational is committed to a judgment factoring through the quotient that removes them, and a failure to factor locates a representational dependence the theory did not admit to having. This is the content of the argument that a symmetry marks superfluous theoretical structure (Ismael and van Fraassen, 2003), stated so a program can test it on a particular case.

4.3 Invariance is an envelope rather than a predicate

Pass and fail are too coarse for discovery. A theory may be invariant under unit changes and not under scale changes, stable across paraphrases and not across languages, transferable to low-stakes contexts and not to consequential ones.

Definition 2 (Invariance envelope). *For a theory or judgment procedure J and a universe U of candidate transformations, the invariance envelope is*

$$\mathcal{E}(J) = \{ \tau \in U : J(x) \sim_Y J(\tau x) \text{ over the declared test domain} \}. \quad (3)$$

The test domain must be reported, because an empirical envelope is always relative to sampled cases, measurement tolerance, and statistical power. The formal object is an idealization and empirical work estimates it. Treating invariance as a matter of degree and range is Woodward’s move (2003), and the set of routes across which a result holds is Wimsatt’s (1981). The envelope adds one requirement to both, that U be written down before the test rather than recovered from it afterwards.

Definition 3 (Comparative fundamentality). *Let J_1 and J_2 be candidate descriptions with comparable empirical adequacy and without compensating increases in unconstrained complexity. Then J_1 is more invariantly fundamental than J_2 relative to U if $\mathcal{E}(J_2) \subseteq \mathcal{E}(J_1)$.*

This is comparative and local by design. It does not identify metaphysical fundamentality with invariance, and it is not offered as a contribution to the metaphysics of fundamentality, where the relations at issue are grounding and ontological priority rather than survival under redescription. It says only that when two descriptions account for the same phenomena, the one surviving more meaning-preserving changes has discarded more representational contingency. A reader who holds Nozick’s (2001) account of objectivity may read the definition as that account restricted to a declared U and made decidable for a particular pair.

Remark 1. The complexity qualification blocks a vacuous victory in which invariance is bought by adding a correction for every transformation. The discipline therefore pairs envelope breadth with parsimony, preregistration, and prospective prediction. Without the qualification, comparative fundamentality would reward exactly the ad hoc manoeuvre the robustness literature identifies as its own main risk.

Two properties of this ordering are worth stating, because both limit it.

Proposition 2 (Envelope inclusion orders classes, not theories). *Fix U and write $J_1 \succeq_U J_2$ when $\mathcal{E}(J_2) \subseteq \mathcal{E}(J_1)$. Then $\succeq_U$ is reflexive and transitive, and $J_1 \succeq_U J_2$ together with $J_2 \succeq_U J_1$ gives $\mathcal{E}(J_1) = \mathcal{E}(J_2)$ rather than $J_1 = J_2$. So $\succeq_U$ is a preorder on judgments and a partial order on the classes of judgments sharing an envelope. If U and the declared test domain are finite and $\sim_Y$ is decidable, then $\mathcal{E}(J)$ is computable in $|U|$ times the size of that domain, and $\succeq_U$ is decidable.*

Proof. Reflexivity and transitivity are those of set inclusion. Mutual inclusion gives equality of envelopes, which does not determine the judgment, since two judgments can survive the same transformations and still differ on cases no transformation in U reaches. Decidability is by exhaustion over U and the domain. $\square$

The consequence limits Definition 3 rather than supporting it. Envelope inclusion cannot separate two descriptions with the same envelope, so it does not rank theories but ranks what theories have discarded. A programme claiming one theory more invariantly fundamental than another therefore owes a transformation lying in the difference of the two envelopes, and a claim of strictness with no such transformation exhibited is not one this definition supports.

4.4 A failure says more when it is reduced to its smallest witness

A radar return is useful because it localizes. The analogue for theories is a counterexample small enough to name a cause.

Definition 4 (Boundary witness). *A boundary witness for an invariance claim is a transformation or intervention τ^* such that $J(x) \not\sim_Y J(\tau^*x)$ for some x , while no strictly simpler admissible predecessor under a declared complexity ordering produces the violation.*

For a finite transformation grammar, breadth-first search identifies a minimal violating composition under word length (Bond, 2026c). The move is standard in software debugging, where delta debugging narrows a failing input to a minimal failure-inducing core (Zeller and Hildebrandt, 2002), and the methodological content is identical. A reduced failure names a component and an unreduced one does not. For continuous parameters the boundary is estimated as

$$t^* = \inf\{t : D(J(x), J(T_t x)) > \epsilon\}, \quad (4)$$

with D an output discrepancy and ϵ a tolerance fixed in advance. A failed invariance test then measures where a degree of freedom supposed to be irrelevant starts to matter.

Definition 4 is not satisfiable under every declaration, and the condition is worth recording because it is what the complexity ordering is for.

Proposition 3 (Witnesses exist when the ordering is well-founded). *Let $\prec$ be the declared complexity ordering on U , and let $V \subseteq U$ be the non-empty set of transformations under which J is not invariant. A boundary witness exists if and only if $\prec$ is well-founded on V . A witness is unique when $\prec$ is total on V , and need not be otherwise.*

Proof. If $\prec$ is well-founded on V then V has a $\prec$ -minimal element, which satisfies Definition 4. If it is not, there is an infinite strictly descending chain in V , so no element of V is minimal and no witness exists. Two $\prec$ -incomparable minimal elements of V are both witnesses, which gives non-uniqueness, and totality removes incomparability. $\square$

This is why a declaration carries a grading and not only a membership list. A universe presented as an unstructured set has no minimal elements to find, so the localizing work that separates this account from a pass and fail verdict is unavailable to it, and a programme reporting witnesses without having declared an ordering has reported counterexamples under another name.

4.5 Narrowing the admissible class has to be shown to cost something

The standing objection to any invariance-based account is that the theorist picks the class. Section 2 credited Nozick (2001) with raising it and leaving it open, and an account that answered it by declaring the class and stopping there would deserve the same verdict, because a recorded arbitrary choice is still arbitrary. The point bites hardest at Definition 3, which makes comparative fundamentality relative to U . If nothing constrains U , that definition ranks nothing, and the apparatus above is bookkeeping attached to a free parameter.

Four constraints are already carried by the definitions. A fifth is a norm this paper proposes, and it is the one that does the work the others leave undone.

The class is fixed by the claim rather than chosen beside it. An admissible transformation is licensed by the philosophical commitment under test, so $\mathcal{T}$ is not a second choice made after the claim but a restatement of it. A claim that renaming does not matter fixes its class as soon as renaming is defined. The theorist's freedom is therefore freedom over what to claim, which

is the ordinary condition of inquiry rather than a defect peculiar to this method. What changes is that the two move together. Narrowing $\mathcal{T}$ to save a result narrows the claim by exactly as much, and because the claim is registered before the test, the narrowing is a visible amendment rather than a silent one.

Declaring a transformation declares what it generates. Definition 1 quantifies over the closure $\langle \mathcal{T} \rangle$, not over the declared generators. A class gerrymandered to exclude an awkward case must therefore be closed under composition while still excluding it, which is a strong structural demand and usually an impossible one. Most convenient exclusions are not closed, and a reader can check closure without rerunning anything.

Breadth is bought from discrimination. Principle 1's non-triviality requirement means survival across a wider class is not free. A judgment can always be made invariant under more by discriminating less, and in the limit a constant judgment survives everything and says nothing. The quantity that Definition 3 compares is therefore envelope breadth at fixed discriminative adequacy, which is why that definition carries the comparable-adequacy clause. The clause is not a technical caveat. It is what stops the ranking from rewarding theories that have purchased invariance by saying less.

A class without an ordering supports no witness. Definition 4 requires a declared complexity ordering, since minimality is defined against it. A universe presented as an unstructured set supports no boundary witness, hence no boundary claim, hence none of the localizing work that distinguishes this account from a pass and fail verdict. Requiring witnesses therefore constrains the shape of U and not only its membership, and a registry whose families are all of equal rank has declared no ordering and can claim no minimality.

A restriction has to bind. The four constraints above make a class costly to gerrymander and none of them makes it impossible, because a programme can still declare a narrow class from the start and claim only what that class supports. The remedy is a norm on restrictions. A programme that restricts its class or its domain must show that the restriction does work, which means the claim must fail, or come measurably closer to failing, outside the restriction. Where the claim holds equally well outside the declared scope, the scope was not a boundary of the claim but a convenience of the report, and the claim should be stated without it.

The norm is mechanical, which is its point. A declared scope carries two obligations rather than one. The claim must hold inside it, and the claim must be worse outside it by a margin fixed in advance. Section 7.5 reports a case in which both were registered before the run and both were graded. The second obligation was met, the out-of-scope error reading 1.19 against an in-scope limit of 0.64, and had it not been met the registration said in advance that the claim would fail as stated. That clause is what stops a scope from being a way of dropping the cases that do not work, and a programme unwilling to write it is telling a reader something.

What this does not do is make U objective. Nothing here recovers a class that the world hands over, and outside mathematics and physics no such class is on offer. What it does is convert an unconstrained choice into a constrained and visible one. The class must follow from the claim, must be closed, must be paid for in discrimination, must be ordered, and must bind where it is narrowed. A disagreement about U then becomes a disagreement about which of those five a rival registry fails, which is a question with an answer, where the disagreement it replaces was a difference of intuition about what ought not to matter.

One consequence should be stated because it limits Definition 3 rather than supporting it. Envelope inclusion compares two descriptions only when both are evaluated against the same declared U . Two programmes with different registries are not ranked by that definition at all, and a claim that one theory is more invariantly fundamental than another is a claim made inside a shared declaration. Comparative fundamentality is accordingly a relation on theories and a declaration,

written $\mathcal{E}_U$ where the subscript matters, and a bare ranking should be read as one whose U has gone unstated.

5 The protocol

The discipline organizes work as a sequence of increasingly severe tests, severe in Mayo’s sense (2018). The objective is not to maximize the phenomena a theory can be made to fit after the fact. It is to find the smallest structure that still predicts after everything the theory says should not matter has been changed.

5.1 Formalize the commitment

Replace broad language, such as objective, rational, same case, irrelevant framing, and fundamental, with explicit mathematical commitments. These may be equivalence relations, group actions, order restrictions, metric identities, conservation conditions, commutative diagrams, or conditional independences. Formalization must separate four statuses, which are definitions, modelling assumptions, derived propositions, and empirical hypotheses. The common error in interdisciplinary work is treating a convenient representation as an observed law, and the separation prevents it.

5.2 Derive consequences that divide rivals

A good probe is not one under which the theory could fail. It is one under which competing theories diverge. Candidate consequences should be ranked by expected discriminatory value, and invariance where rivals predict sensitivity, sign reversals, boundary locations, low-rank constraints, monotonicity, and forbidden orderings are the useful kinds. The strongest tests are often consequences the theorist would not have chosen, and they are valuable because the formal structure forces them.

5.3 Search for low-complexity structure

The discovery engine may be human reasoning, symbolic regression, equation discovery, structure learning, or a hybrid. Symbolic regression is attractive because it searches for concise relations rather than only for predictive black boxes (Makke and Chawla, 2024), recent work applies it to interpretable model discovery under physical constraints (Hou et al., 2024), and broader automated-science systems combine representation learning, equation discovery, experiment selection, and closed-loop autonomy (Kramer et al., 2026). A search objective can be written

$$\min_{F \in \mathcal{H}} L(F) + \lambda C(F) + \mu I(F), \quad (5)$$

with L a predictive loss, C a structural complexity, and I a penalty on instability under transformations the programme has declared irrelevant. This is not proposed as a unique objective. It states the preference for descriptions that are accurate, simple, and invariant together.

5.4 Attack the representation

Subject a candidate law to transformations preserving information while changing coordinates or presentation. Basis rotations, unit changes, equivalent encodings, feature permutations, redundant embeddings, alternative encoders, and controlled compressions are the usual instruments. Structure

Level	Question
Fit	Can the structure describe the observed cases?
Perturbation	Does it survive local parameter and data perturbation?
Reparameterization	Does it survive information-preserving changes of coordinates?
Holdout	Does it predict unseen cases from the same population?
Transfer	Does the same structure survive a different task or population?
Cross-domain	Does it organize qualitatively different phenomena?
Adversarial probe	Does it survive a deliberately chosen discriminator?
Boundary map	Where does it cease to apply, and which assumption breaks?
Replication	Can an independent group reproduce the chain?

Table 1: The ladder of probes. A result earns only the rung it has reached, and the rung a programme reports is the first thing a reader should check, since every rung above Holdout is one a well-fitted artifact can still fail.

disappearing under an information-preserving change of coordinates may belong to the representation rather than to the phenomenon. Contemporary symmetry discovery does the corresponding technical work, enforcing known symmetries, searching for unknown ones, and breaking candidate symmetries only where data require it (Otto et al., 2025), and causal representation learning has been reframed so identifiability runs through preserved invariances rather than one fixed formalism (Yao et al., 2025). The philosophical reading is that a distinction should be kept only if the world forces us to keep it.

5.5 Freeze, then probe what was not used to build it

After discovery, freeze parameters and transformation rules as far as possible and evaluate on unseen cases, populations, scales, or domains. This is where a suggestive reduction becomes evidence of transferable structure. Table 1 states the ladder.

5.6 Reduce failures rather than hide them

A null result may be uninformative. A localized contradiction is not. When an invariance fails, find the smallest witness and ask which component absorbs the change, the candidates being the representation, the metric, the constraints, the resolution budget, the dynamics, and the philosophical equivalence relation itself. This turns falsification from a verdict into cartography. The aim is not to immunize a theory against failure but to make its failures diagnostic.

6 Where this sits among existing methods

Experimental philosophy studies the distribution and stability of judgments elicited by cases, and recent work continues to debate calibration, comprehension, and robustness to error (Knobe and Nichols, 2024; Iriekfe, 2025; Cova and Martinez, 2025). The present discipline shares the commitment that philosophical practice can be interrogated empirically and differs in its unit of analysis, targeting formal consequences of philosophical theories rather than primarily the psychology of philosophers or lay participants. Human judgments may be the dependent variable, and so may physical measurements, algorithmic behaviour, economic choices, or theorem-prover outcomes.

Computational philosophy uses simulation, automated reasoning, and formal modelling to instantiate and extend philosophical research (Thagard, 2024), and this is a narrower proposal inside

that space, distinguished by the closed loop from conjecture to executable test to reduced witness to revision. Formal philosophy supplies the language without which the proposal would be metaphor, and the discipline adds the requirement that a formal claim with observable consequences also specify how those consequences are tested and what would count as a minimal counterexample. Automated scientific discovery has moved from equation discovery toward systems selecting experiments and learning representations (Kramer et al., 2026), and the difference here is the type of prior knowledge compiled, since the starting commitments are philosophical and part of the goal is to find which of them survive contact with the world.

7 The discipline at work

Six instances follow. Each is introduced by the philosophical claim it puts under test, because a case that does not name one is an engineering report. The first is generic and the next three are brief. The fifth ran the protocol of Section 5 to its declared end and is the only case that did. The sixth is the one in which the discipline’s own check found a defect in the records of the programme that wrote it.

Every number in Sections 7.2, 7.5, and 7.6 is read from a sealed record in a public repository, cited where it appears. None is restated from a draft.

7.1 Epistemic invariance as an experimental object

The claim under test is that two descriptions differing only in a meaning-preserving way are the same case, which is an equivalence claim about representations (Bond, 2026c). In its engineering role it is a specification, so a system changing a medical recommendation because laboratory results were reordered violates a declared norm. The discovery role is different.

Suppose a class $\mathcal{T}$ is proposed as semantically irrelevant. An engineer can build the transformation suite, measure judgment discrepancy, and search for minimal witnesses. Three outcomes are possible. Invariance may hold broadly, and the claim that the transformed inputs are the same case gains operational support. Invariance may fail because the model is representation-sensitive while the philosophical equivalence remains compelling, in which case the system is defective against the norm. Or invariance may fail systematically in a way showing the proposed equivalence erased a relevant distinction, in which case the philosophy and not only the implementation should change. The third outcome is the distinctive one, and it is where engineering exposes a hidden philosophical assumption by forcing it to meet a concrete counterexample.

7.2 A campaign that graded four of its own predictions

The claim under test is that evaluation is evaluator-relative in a specific way, the world supplying states, actions, and a consequence map, and the evaluator supplying a metric, an ideal point, and a finite budget that limits the distinctions it can make, with preference and choice derived from those distinctions (Bond, 2026a). Every claim in that programme is labelled proved, defined, posited, or open, and predictions are graded through gates sealed before the data are read. Four questions have run across five sealed registrations.

The first tested a consequence of the representation theorem, that no action is strictly worse than every action whose consequences surround it. The consequence map must be known, so the world was the 1972 American National Election Study, where respondents placed candidates and parties on seven-point issue scales and rated them on feeling thermometers, and the registration was sealed before any rating was read. On the registered pair of scales 489 respondents had a

menu in which the law could bind, 50 percent violated it at least once, against 66 percent when each respondent’s ratings were shuffled among that respondent’s own objects, and no shuffle in 200 reached the observed rate. Five other pairs of scales gave the same ordering. The verdict by the sealed bars was a pass, and the number that matters more than the verdict is the half. The theorem is deterministic for a representable preference, so the law holds in this population as a tendency and fails as a law for half the respondents. In the vocabulary of Section 4.4 the population is a boundary witness against the deterministic reading, and what absorbs it is not yet registered, since no error model for integer placements and integer ratings was part of the seal.

The second tested whether the indifference threshold of the induced order is the evaluator’s resolution budget. The evaluator was a language model asked how far a value sits from a target, and three budgets were set independently, the decimals shown, the tokens allowed for the report, and the precision of the weights. On fresh pairs every graded threshold sat at 0.87 to 0.88 of the step the rendering imposed and of the resolution the report length afforded, the largest ratio 1.15 against a tolerance of 1.5 fixed before the seal. The weight precision did not move the threshold at all. Two of three declared budgets are resolution budgets and the third is not, which is the finding, and the registration could have carried only the two that worked. Two instruments tried before the one that ran are kept in the record rather than dropped, and the reduction of their failure generalizes past the case. A pairwise chooser measures its own position bias before it measures a threshold.

The third tested a theorem about several evaluators reading one rank-limited representation, and is reported here because its failure was informative rather than because it passed. The first sealed run found the measured loss 20 to 330 times the second-order prediction, so the formula was not tested and the verdict was indeterminate. The failure was then reduced, as Section 5.6 requires, to two causes. The registered prediction and the registered measurement were not the same object, and the codes sat far outside the range where the quadratic approximation holds. A second sealed run measured the object the formula predicts and found the ratio of measured to predicted loss flat in the perturbation size with median over cells 1.02 to 1.05, while 64 draws per cell left a scatter that missed the registered bar, so that verdict was indeterminate as well. What the two runs establish together is the theorem’s range of validity. Inside it the empirical content reduces to whether the loss is quadratic with the recovered operator, which holds. Outside it an ordering claim survives and a claim about private optima does not. The perturbation ladder is equation (4) executed.

The fourth tested identification, that an evaluator’s choices recover the metric up to scale and the ideal up to the metric’s kernel, with the design bound that the consequences must span the space affinely. Two pilots before the seal caught two wrong drafts of the estimator and showed the drafted tolerances were empty, so the bars were restated on medians relative to each evaluator’s chance level. The sealed run passed in all twelve cells on a fresh seed, and at or below the design bound, off the span of a battery confined to a subspace, and in the kernel of a singular metric, nothing was recovered. The negative claims, what choices cannot reveal, came out as sharp as the positive one, and each would have been a witness against the theorem had it failed.

This theory suits the discipline because it makes representation choice explicit. Under an invertible change of basis $z' = Az$ a quadratic metric transforms as

$$G' = A^{-T}GA^{-1}, \quad (6)$$

so that

$$z'^T G' z' = z^T G z, \quad (7)$$

and the evaluative distance is invariant while the coordinates are not. Which properties of evaluation survive changes of representation, encoder, dimensionality, and task is then the question the

campaign asks gate by gate.

7.3 A published invariance that failed on independent data

The claim under test is that a decision geometry fitted in one setting describes decisions in another, and the invariance it entails is that a quantity the fitted model treats as inactive cannot matter. This is the cleanest case in the paper, because the invariance was forced by the model rather than chosen for testing.

A geometric model of economic behaviour accepted by a systems journal represents alternatives in a nine-dimensional decision space and fits a diagonal Mahalanobis metric (Bond, 2026e). Structural search selects social impact, virtue and identity, and epistemic status as active dimensions on the reported benchmark, and the monetary dimension is inactive in the selected low-stakes calibration. Under the share-normalized encoding, positive rescaling of all monetary stakes should therefore leave predicted offers and rejections unchanged, which is a sharp stake-scaling invariance that nobody chose to claim.

Tested on individual-level high-stakes ultimatum data that played no part in fitting or selection, offers and rejection rates changed substantially with stake, and stake retained predictive power after conditioning on offer share. The invariance fails. Under a confirmatory reading this is a negative result. Under the present one it is a boundary measurement, because the contradiction identifies a specific missing quantity, share normalization having removed consequential monetary magnitude. An income-relative monetary coordinate is then a new testable hypothesis, that decisions depend on stake relative to local income rather than on nominal stake. The route runs from geometry to invariance to independent probe to boundary witness to a new representation. A radar return is not a failure. It is information about where the theory stops describing the terrain.

7.4 Symbolic search that reports when it loses

The claim under test is that simplicity is evidence, which the case qualifies rather than supports. Theory Radar discovers compact classifiers from tabular data and evaluates whether they can fairly replace ensemble models on the dataset at hand (Bond, 2026d). Its stated value lies in the comparison rather than in symbolic formulas winning, and a formula is accepted only if it survives fair held-out evaluation. Where the boundary is intrinsically high-dimensional the ensemble wins, and that negative result is informative because it identifies a domain in which compression to the searched grammar loses predictive structure. The natural extension scores candidates for stability under admissible representation changes as well as for accuracy and syntactic size, which turns symbolic search into a way of finding simple formulas whose content survives transformations that should not matter.

7.5 Nine registrations and one law

The claim under test is that a property attributed to a body of data is partly a property of the reader, so that a law about it must be stated in the reader’s variables or not at all. The cases above each exercise part of the loop. This line exercises all of it, nine times under sealed registrations, and is the only case in which the protocol ran to its declared end (Bond, 2026b). It also measures what the protocol costs, since three of the nine registrations failed.

A hub is a row of a corpus appearing in many other rows’ nearest-neighbour lists, and the amount of hubness is summarized against what a random assignment of the same retrieval slots would reach, so that one is chance. The observer is a linear read whose spectrum falls as a power of the index, running from a pure rotation to a reader that sees one direction. Earlier gates had

shown that a corpus’s hubs are made by the reader as much as by the corpus, and the question was whether the amount is lawful in the reader’s own variables.

The family of laws was declared once and kept, linear in at most three features from a fixed pool measured on the observed data. The search ranks candidates by error on held-out worlds under unseen observer spectra, the pilot freezes the best, the registration is sealed by renaming the file with the frozen law inside it, the run uses a fresh seed, and a sealed grader grades. Every bar is a multiple of the frozen law’s own pooled pilot error, fixed before the seal, and a competitor beating the law pooled on the transfer groups is a failure by a clause written in advance.

Four registrations changed the discovery set and kept everything else, and the law changed with the set each time. A law found on Gaussian clouds failed off Gaussians. A law found on seven shape families missed the real embeddings, whose intrinsic dimension sat far under their spectral dimension, because on full-dimensional families those two variables duplicate each other and the search had dropped one. Adding manifold worlds made the search take it back, and the error on the missed corpus fell from 5.3 to 2.5 to 1.0 across three gates. Two further registrations attacked the family rather than the set, and both found the same limit, the largest error of every law sitting where a law linear in any transform of the summary statistic cannot turn over. The record said the family, and not the set, was the limit.

The seventh changed the target and nothing else. On the same worlds and readers the search froze a law with pooled pilot error 0.328, a factor of 1.39, against 1.135 for the nominal formula and 1.598 for chance, and on the run it held the heavy tails, the highest dimensions, fresh real slices, and the largest samples at once, beating both competitors pooled and inside every group for the first time in the line. Same family, same pool, same search, same worlds. The target had been the limit all along. The eighth asked whether the new target takes coverage without paying for it, and it does not, buying the highest-dimensional worlds and paying on the heavy tails at worse than chance, so the earlier law won pooled and the gate failed by its own clause.

An exploration then asked whether any shape of the declared family holds both sides at once. Twelve configurations were scored and none passed, and the exploration named the mechanism instead. The law’s concentration term is a reciprocal, it diverges where that concentration is small, and it predicts an excess near zero against a truth of 1.4. Removing the reciprocal loses the real corpora, which the reciprocal is what fits. That is a boundary of the family’s shape and not a missing world, and it is Definition 4 carried out on a declared complexity ordering.

The ninth registered that boundary as the claim and searched nothing, declaring a scope from the collected rows before the pilot. It is the worked case for Section 4.5, because it carried both obligations a declared scope incurs. The law was claimed inside the scope within the earlier bars, and its pooled error outside the scope was required to exceed the in-scope limit, failing the claim as stated otherwise. The registration also named in advance the one in-scope cell it expected to lose, with the reason, which is that the law carries no term in sample size. The run held every bar but that one. All eighteen unseen worlds sat inside the limit on their in-scope rows, pooled 0.31 against a limit of 0.48, the law beat both competitors in every group at 0.34 against 1.63 and 1.64, and the out-of-scope rows erred at 1.19 against 0.64, which is the second obligation holding. The named cell read 0.67 against 0.64. The verdict was indeterminate, by the named cell and no other.

What stands is one law with one declared scope and two recorded boundaries, holding on every family of world the line built, on corpus slices it never saw, and at four times its discovery sample. Three properties are visible in the record rather than asserted. The bars never moved, and three registrations failed on clauses their own authors had written. Two pilots froze a law that trailed a competitor on the pilot’s own rows, and both were sealed and run as registered with that fact recorded, because the transfer groups are the test and the pilot is not. And when a registration that would pass was wanted, the only route the record allowed was a change of claim with its risk

named in advance, which is the route Section 4.5 requires and the reason that section’s norm is not an aspiration.

7.6 The check found a defect its own authors had not

The preceding cases share a weakness a reader should hold against them. Every one is the work of the people proposing the discipline, so each is a report of a method applied by its authors to their own theories. The following episode is the nearest thing to independent evidence the programme can currently offer, because in it the method contradicted the people running it.

The four record types the protocol needs were implemented as machine-readable files, among them a transformation registry declaring, per track, the admissible families with the date each was declared. A checker was then written from Definition 1, testing that every registered test names a family the registry carries. Run against the programme’s own registries for the first time, it reported ten tests, entered across a single day of gate records, citing three families the registry did not contain.

The finding was bookkeeping rather than a priority violation, and establishing which took evidence rather than memory. Each of the three families was declared in its own gate’s sealed registration before the corresponding run, which is checkable in two ways that do not rely on anybody’s recollection. Each seal commit is an ancestor of the commit that added that gate’s results, and each sealed file still hashes to the value the track document recorded. The declarations existed and preceded their tests, and only the registry was late.

Two things about the episode bear on the argument. The first is that the defect had been visible for two days in a published artifact and had not been seen. The generator rendering each entry’s invariance envelope prints a transformation identifier followed by the sentence describing the family, so an undeclared identifier printed an empty sentence, thirty times across four entries and the pages built from them. Reading had not caught it. A check derived from a definition in this paper caught it on first execution.

The second is what the repair had to look like. Adding the three families with the current date would have made the registry pass and would have misdated the declarations. Adding them with the declaration date alone would have concealed that the registry was late. Deleting the offending tests would have removed the record of tests that ran, which is the file drawer the discipline exists to prevent. The rows were therefore entered carrying both dates and a pointer to the sealed artifact, so the registry now says, in a form a stranger can check, that these three declarations were made on one date and recorded on another. The specification was amended to require exactly this, and to state the condition under which a late row is a repair rather than a forgery, which is that the declaration it points at can be shown to precede the result.

We report this because it is the only case here in which the discipline cost its authors something, and because it illustrates the narrow claim of Section 2. No new criterion of invariance was involved. What found the defect was the requirement that the admissible class be a declared artifact rather than a sentence, and what made the repair something other than cosmetic was the requirement that the record show its own history.

8 The fundamental structure conjecture

The cases motivate a conjecture rather than a conclusion.

Conjecture 1 (Fundamental structure, as a hypothesis about yield). *In domains where the protocol of Section 5 can be run, searching for what survives admissible changes of representation, scale,*

population, and task will more often than not isolate a structure of lower complexity than the representation it acts on, and one reaching at least the Transfer rung of Table 1.

It is stated as a hypothesis about what the method yields rather than as a claim about the world, because the claim about the world, that many phenomena admit such structures, cannot fail and so is not worth asserting. As stated it can fail. What would fail it is a programme that runs the protocol across several unrelated domains, declares its classes in advance, and finds in each that nothing survives past the Holdout rung. That outcome is available, it would be reportable under the same discipline as any other negative, and nothing reported here rules it out.

Three qualifications stand whichever way the hypothesis is read. Low dimensionality is not guaranteed, and some phenomena may be irreducibly complex at the scale of interest, the cases in which ensembles beat compact formulas being exactly the evidence a serious programme preserves, along with the shape boundary of Section 7.5 where the declared family had no form holding both sides. Invariance is always relative to a transformation class, so what survives translation may fail under a change of scale and what survives within a culture may fail across cultures. And invariance does not imply ontology, being evidence that a structure depends less on chosen descriptions rather than proof that it is metaphysically ultimate.

The conjecture nonetheless supplies a search strategy. Begin with a rich representation, identify a compact structure, transform the representation, transfer the structure, and keep what survives. Repeated quotienting of representational contingencies can expose an empirical core that becomes progressively harder to remove without losing predictive power.

9 Elegance, rigour, and the risk of beautiful error

The discipline gives a precise role to two virtues. Elegance pushes toward fewer primitives, reusable structures, and compression across apparently unrelated phenomena. Rigour prevents elegance from becoming an aesthetic substitute for evidence.

The danger is plain. A mathematically beautiful model can be wrong, and the more beautiful it is the easier it is to rationalize away contrary evidence. Generating hostile tests therefore belongs to the theory rather than to a later optional exercise, and a mature research artifact should carry a claim ledger, a separation of definitions from measured claims, a registry of admissible transformations, preregistered gates, holdout data, ablations, and explicit statements of what would force revision. Section 7.5 shows what the last of those costs when it is enforced, which is three failed registrations out of nine.

This is consistent with a pragmatist stance. The reality of a proposed structure is not secured by metaphysical assertion. It earns standing by generating correct, transferable predictions and by surviving increasingly severe attempts to remove representational artifacts.

10 A research programme

Five workstreams follow.

A calculus of philosophical invariants. Develop a common formal language for equivalence claims across epistemology, decision theory, ethics, philosophy of mind, and philosophy of science. Each application specifies the representation space, the output space, the transformation set, the output equivalence, non-degeneracy criteria, uncertainty sets, and witness complexity. The immediate deliverable is a taxonomy of invariance claims, which are exact, approximate, equivariant, conditional, local, distributional, and scale-dependent.

Representation attacks. Construct benchmarks in which the same underlying cases are presented through alternative coordinate systems, embeddings, languages, schemas, and modalities, requiring candidate theories to survive invertible transformations and to degrade predictably under lossy ones. This connects the discipline to representation learning and symmetry discovery.

Low-rank evaluation. Test whether evaluator geometry inferred in one domain transfers to another under independently determined representations. The decisive experiments identify an evaluator in one domain, freeze the structure, and predict choices in two others with minimal target-specific adjustment.

Automated theory search. Extend symbolic search from classification toward general structure discovery with explicit complexity and invariance objectives, with candidate outputs including equations, metric tensors, low-rank subspaces, transformation groups, and boundary conditions, and with discovered structures replayed on untouched test data.

Boundary atlases. Rather than publishing only the domains where a theory works, build atlases of where invariances fail, estimating for each claimed symmetry the boundary in scale, context, population, or representation at which the effect becomes detectable. These maps may be worth more than a list of successes, because they name the variables the next theory needs. The two boundaries of Section 7.5, one in the concentration of a distance and one in sample size, are entries in such an atlas, and each names a term a successor would have to carry.

11 Objections

This is falsificationism with software. The discipline is compatible with falsificationist virtues and is not exhausted by binary refutation. Its distinctive object is the structure of the counterexample, and minimal witnesses, transformation grammars, and boundary estimates turn a failed prediction into information about which claimed equivalence stopped holding. The method is closer to debugging a formal architecture than to recording a rejected hypothesis (Zeller and Hildebrandt, 2002).

Invariance is not truth. Correct. A false theory can be invariant and a true local theory can legitimately fail outside its domain. van Fraassen (1989) presses the stronger version, that symmetry arguments carry substantive commitments about what counts as a difference, and the present account accepts it by treating the transformation class as declared and contestable rather than discovered. Invariance is therefore comparative evidence against representational arbitrariness and not a truth criterion, and predictive adequacy, independent replication, and domain specification remain necessary.

The transformation class is chosen by the theorist. Correct, and it is the objection this account has to answer rather than concede, since Definition 3 makes fundamentality relative to that class. Section 4.5 is the answer, and declaring the class is not it, because a recorded arbitrary choice is still arbitrary. The class is constrained five ways, by following from the claim rather than being chosen beside it, by being closed under composition, by being paid for in discriminative adequacy, by carrying the ordering a witness needs, and by the requirement that any narrowing of it be shown to bind. None of the five makes the class objective, and together they turn a free parameter into a constrained one whose failures can be named, so that a disagreement about the class becomes a question of which of the five a rival declaration fails. In the programmes reported here the registry is a file, it is sealed before the runs that test it, and Section 7.5 is a case in which the declared family and not the data turned out to be the limit.

Philosophical claims are often normative. Normative claims are not settled by observing behaviour, and the discipline does not infer an ought from an is. It tests the operational

consequences of a normative principle once the principle is formalized, which means whether an implementation respects declared equivalences, whether a proposed standard is internally invariant, whether a policy rule produces representation-dependent outcomes, and whether a normative model predicts human judgments when offered as a descriptive theory.

Search will overfit. Automated search creates severe multiplicity and researcher degrees of freedom, so the discipline requires separation of discovery from evaluation, fair holdout replay, preregistration where feasible, and independent replication. An expression found after searching millions of candidates is not evidence until its search exposure is accounted for and its performance survives untouched data. Section 7.5 is the worked case, the search having run nine times over one declared family with every bar a multiple of a pilot error fixed before the seal, and the two gates whose frozen law trailed a competitor on the pilot's own rows run as registered rather than re-searched.

The evidence is the proponents' own work. This is the objection we think strongest. Every case in Section 7 comes from one programme, and a method applied by its authors to their own theories is not independently tested, however sealed its registrations. Three things mitigate and none removes it. The registrations, data, graders, and run logs are public, so the verdicts are checkable even though they are not independent. The record retains outcomes that embarrass the programme, among them three failed registrations, two indeterminate gates, a published invariance that failed on external data, and a population in which a deterministic theorem holds for half the respondents. And Section 7.6 is a case in which the discipline's own check overruled its authors. What is missing is a second programme, run by other people, and until that exists the claims here are claims about a method demonstrated once.

12 Engineering as an epistemic instrument

The phrase philosophy engineering can suggest that philosophy supplies requirements and engineering implements them. The relationship proposed here is stronger. Engineering exposes the hidden commitments of a philosophy by forcing them into interfaces, data structures, registries, test cases, and failure modes, and ambiguities surviving prose become hard to ignore when a system must decide whether two cases belong to the same equivalence class.

This changes the epistemic role of implementation. A program is not only a demonstration that a theory can be instantiated. It can be a measurement instrument for the theory. Canonicalizers test claims about sameness. Equivariant architectures test claims about symmetry. Symbolic search tests claims about compressibility. Cross-domain transfer tests claims about generality. Minimal witnesses test claims about irrelevance. Boundary maps test claims about scope.

The resulting practice is neither armchair philosophy nor ordinary empirical science. It is a disciplined traffic between conceptual analysis and experiment. Philosophy proposes distinctions, mathematics makes them precise, engineering makes them executable, experiment pressures them, and philosophy must then explain the returns.

The operational sense of fundamental that follows is modest. Whatever survives repeated removal of representational contingencies, while preserving discriminative and predictive adequacy, is a better candidate for fundamental structure than what disappears under harmless redescription. The search can fail, and may reveal no compact invariant core. That too is a substantive result.

13 Conclusion

The idea that what survives admissible transformation is a better candidate for the real has been stated four times already, in geometry, in robustness analysis, in the theory of causal explanation, and in structural realism. This paper does not improve on those statements. It argues that the idea has been held in a form that cannot be audited, and that the three things a prose statement leaves in the claimant's hands, the admissible class, the reduction of the failure, and the revision, can be made into declared artifacts a checker refuses when they are absent.

What follows is a discipline whose methodology is six verbs. Formalize, derive, search, probe, witness, revise. We have reported it running on four graded predictions of one theory, on a published invariance that failed on independent data, and on a discovery line that asked one question nine times and ended with one law, one declared scope, and two boundaries. We have also reported the case in which the discipline's own check found a defect in the records of the people who wrote it, which is the only evidence here not resting on our good faith.

The deepest promise is not that philosophy is replaced by engineering, nor that every profound question has a computational answer. It is that some philosophical propositions can be made precise enough to become experimentally vulnerable. Once that happens, implementation stops being application and becomes inquiry.

The guiding question is therefore not whether the fundamental theory has been found. It is more disciplined. What remains after every distinction that should not matter has been transformed away, every compact structure has been attacked on unseen data, and every failure has been reduced to the smallest witness available? Whatever survives that has earned the right to be taken seriously as fundamental.

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The author used a large language model assistant for source preparation, for the prior-art search reported in Section 2, for drafting Sections 7.5 and 7.6 from the sealed campaign records, and for editing. All claims, verdicts, and numbers were checked against the records cited in each section, and the author is responsible for the content.

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